AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 12 · STUDENT LESSON

Speech and Audio Systems

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Keep the source. Trace each stage. Locate the first supported mismatch.

Diagnose speech-pipeline errors without treating fluent output or absent recordings as verification.

Keep one habit: Stage contracts → known phrase trace → separate unresolved conflict → four complete cards → independent calculation.

MAIN LESSON · SECTION 1

At which speech stage did the first supported mismatch occur?

> Trace the source, transcript, interpretation, calculation and spoken result separately.

Your learning goals

I can explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall separate criteria for separate stages

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Visual Composition: object presence, spatial relation and arrow meaning need separate checks. In a speech pipeline, recognition, interpretation, calculation and spoken output likewise need their own **evidence**.

Your end goal

By the end, I can separate **speech recognition**, calculation and speech **output** when diagnosing an error.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 12: Speech and Audio Systems

Speech interaction includes **capture**, recognition, interpretation, and response stages.



CREATOR MISSION

Trace errors from noisy audio to wrong action.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

A later stage can correctly process an earlier mistake

> A fluent spoken result does not verify the intended input.

The pipeline and error-propagation idea

A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Use supplied ground truth without claiming a real recogniser run

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use speech and audio systems to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Keep a stage record before assigning blame

> An unavailable source limits what the evidence can diagnose.

Preserve modality, source reference and intermediate outputs

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare conflicting **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

Read the explicit stage contracts

Stage	Output to inspect	Evidence needed for its check
Capture	Recorded audio signal	Available source recording and capture conditions
Recognition	Transcript words	A supplied reference utterance or suitable replay
Interpretation	Operands and operation	Agreement with the confirmed transcript
Calculation	Numeric result	Independent arithmetic using the parsed expression
Speech output	Spoken rendering	Agreement with the intended result text/number

Separate given, derived and unknown facts

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding stage

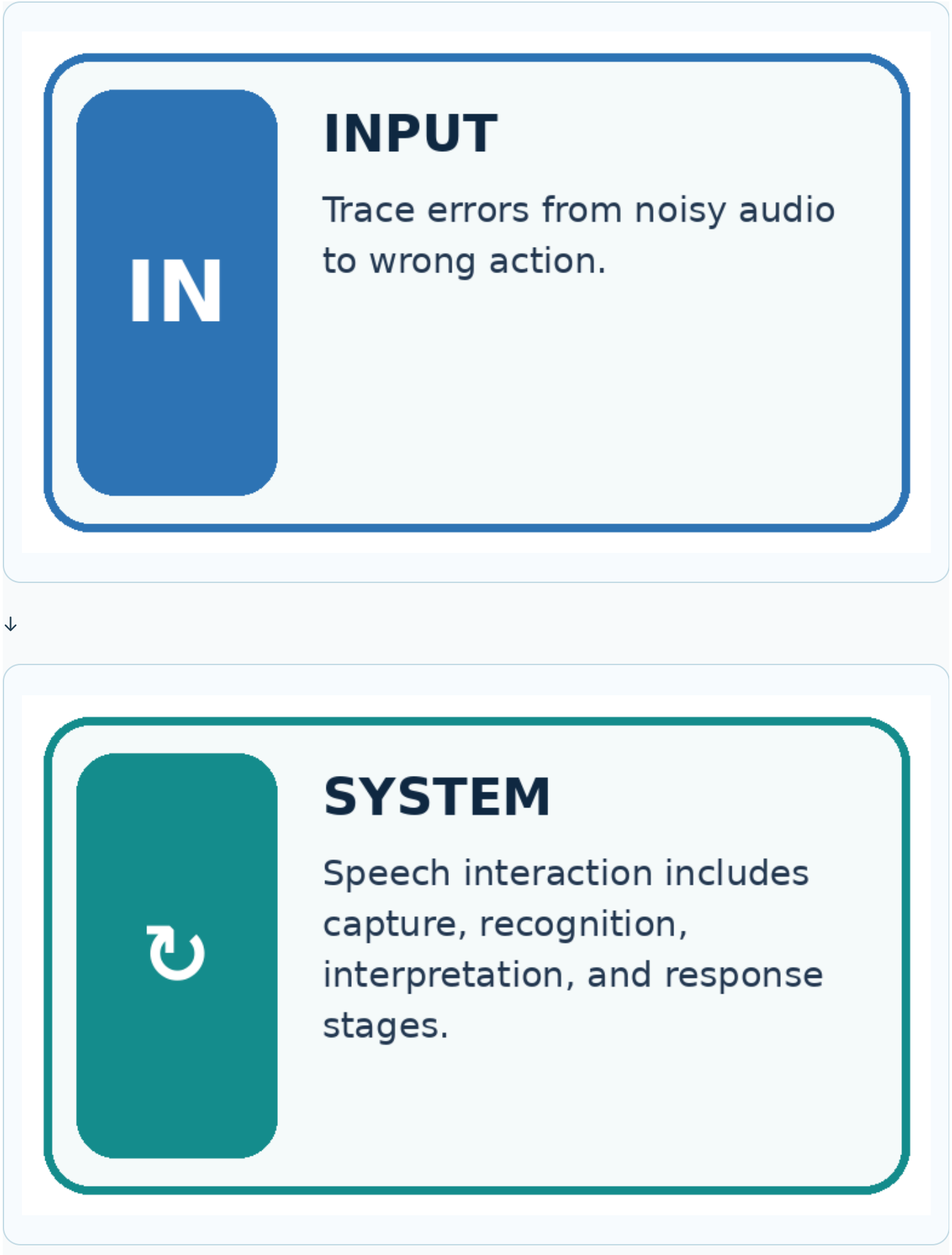
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Trace errors from noisy audio to wrong action.
↓
SYSTEM
Speech interaction includes capture, recognition, interpretation, and response stages.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Speech pipelines and multimodal evidence — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.	Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.	There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

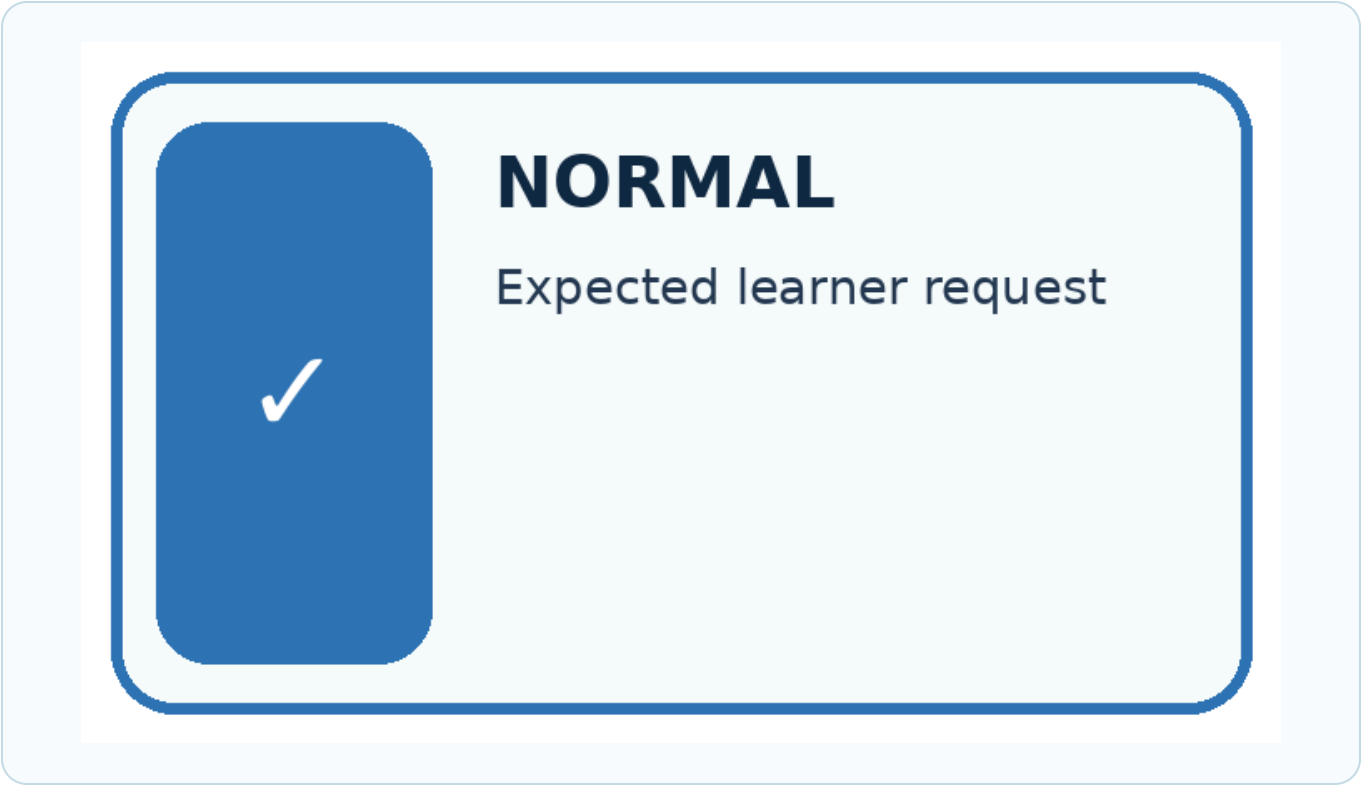
Connect the picture and the explanation

MAIN LESSON · SECTION 4

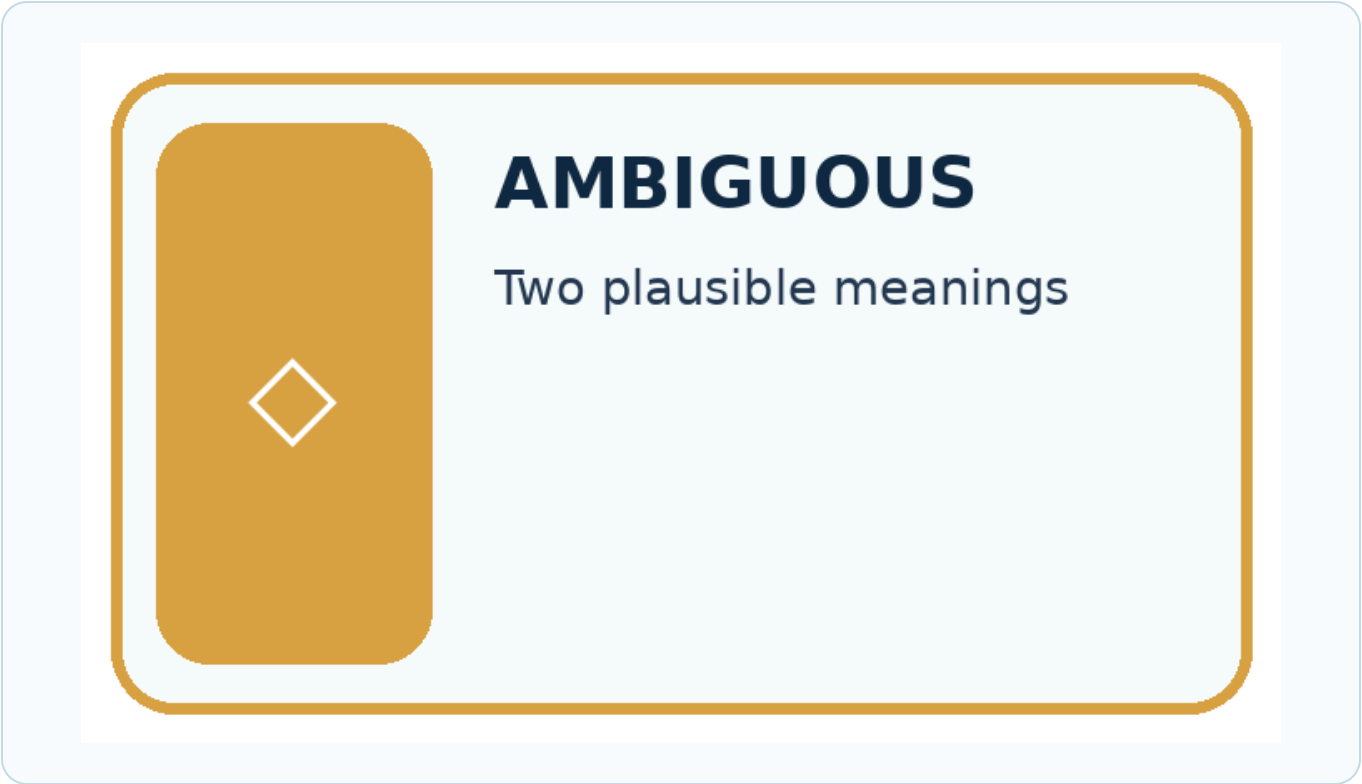
Compare confirmed recognition errors with unresolved source conflicts

> A written record does not automatically settle what an absent recording said.

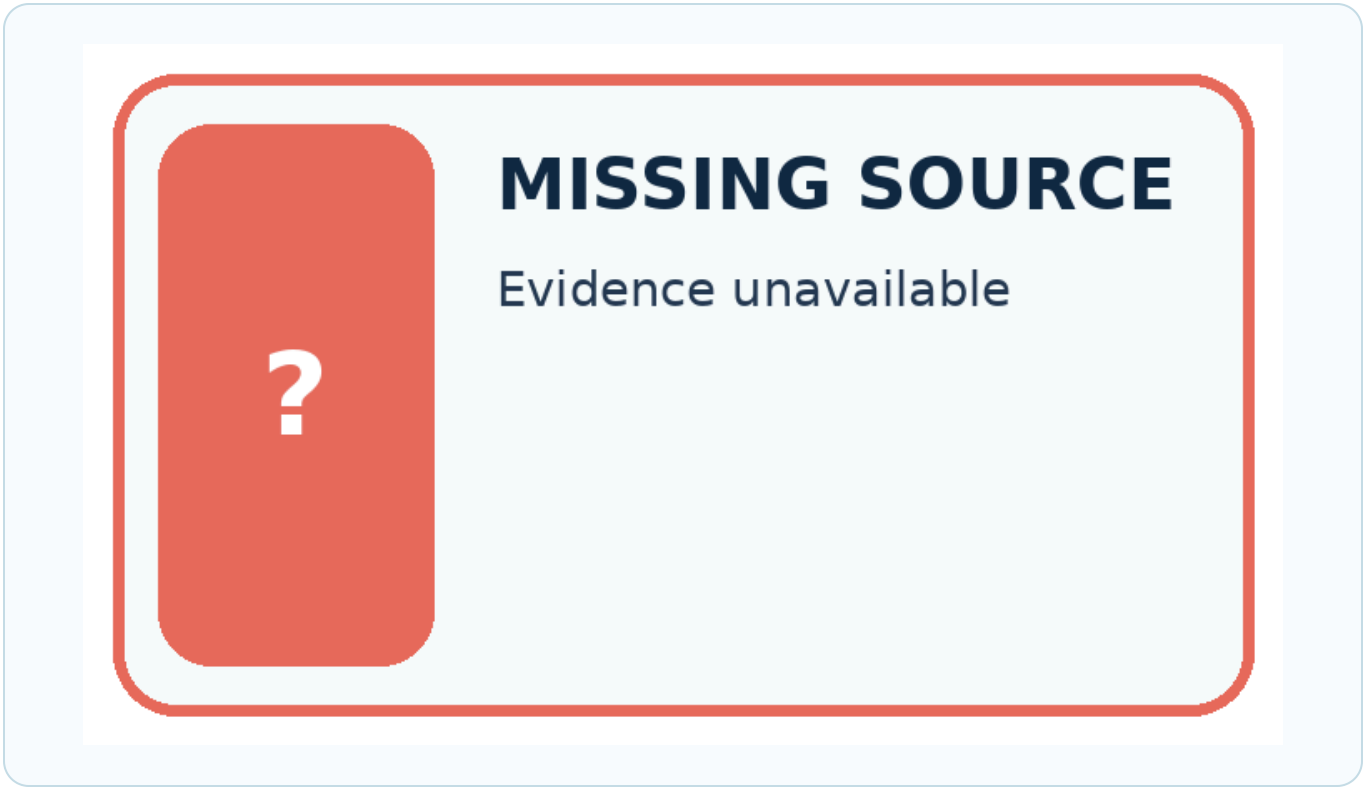
Four pictures with a purpose



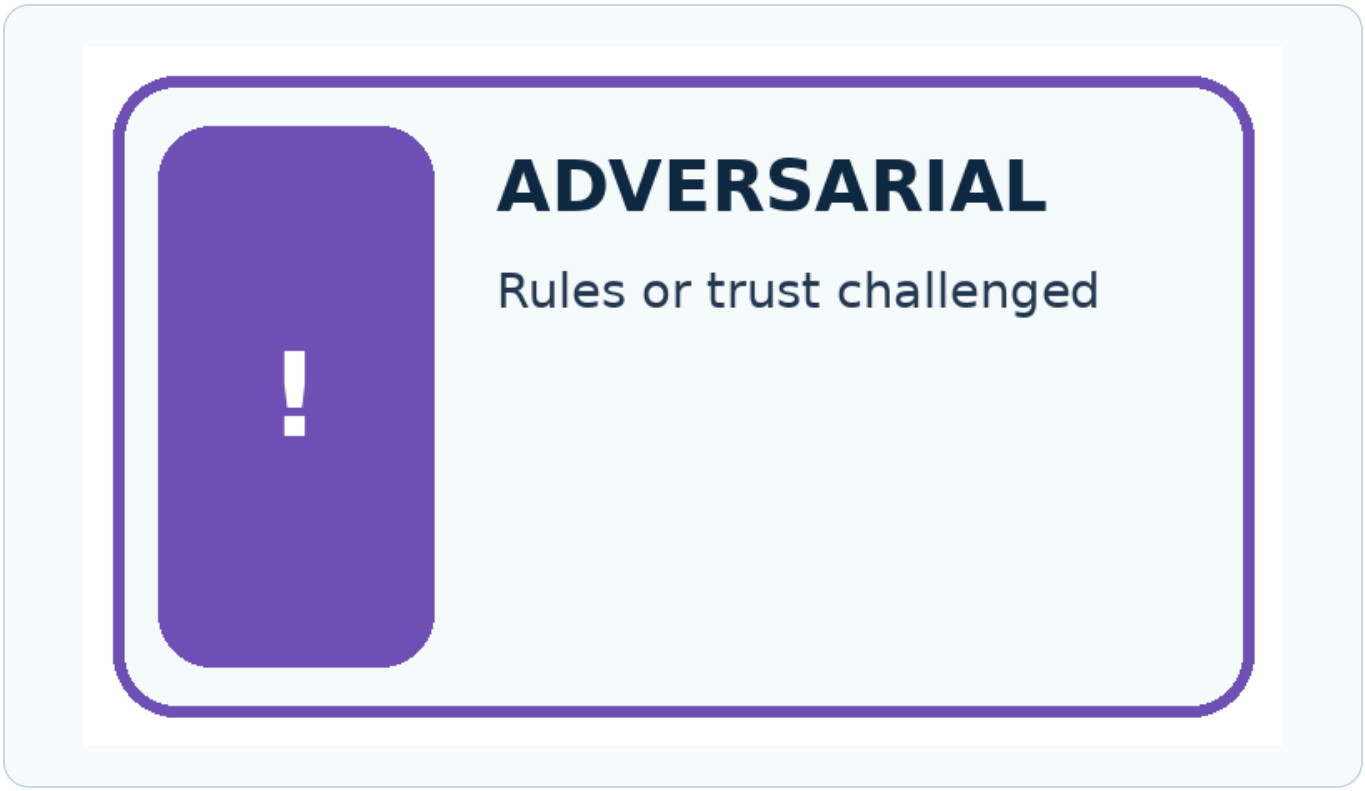
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Correct the transcript to “thirteen plus seven” with all later stages unchanged.

▼ Check after predicting

Prediction: The calculator and spoken result should become 20 under the declared pipeline.

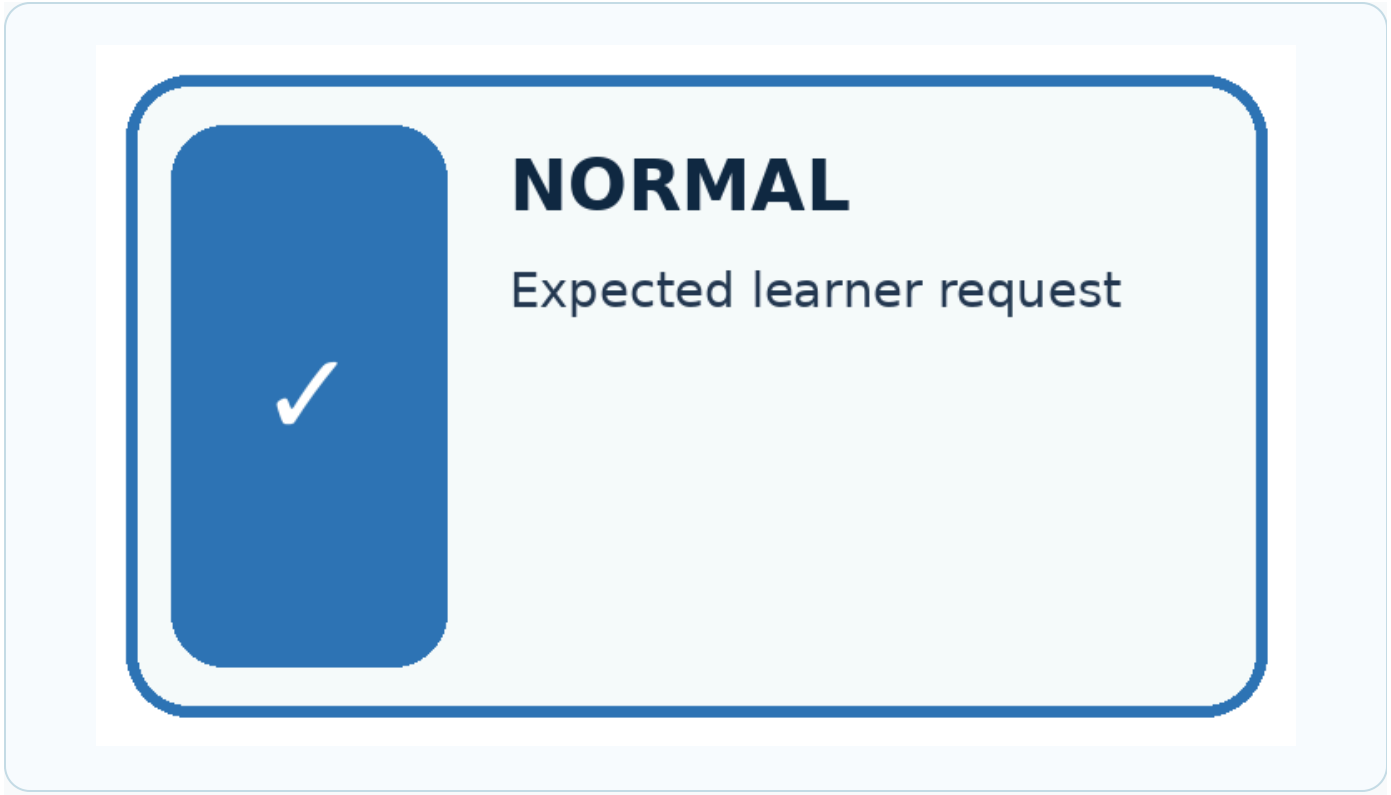
Reason: 13+7=20; the changed **input** removes the earlier recognition error.

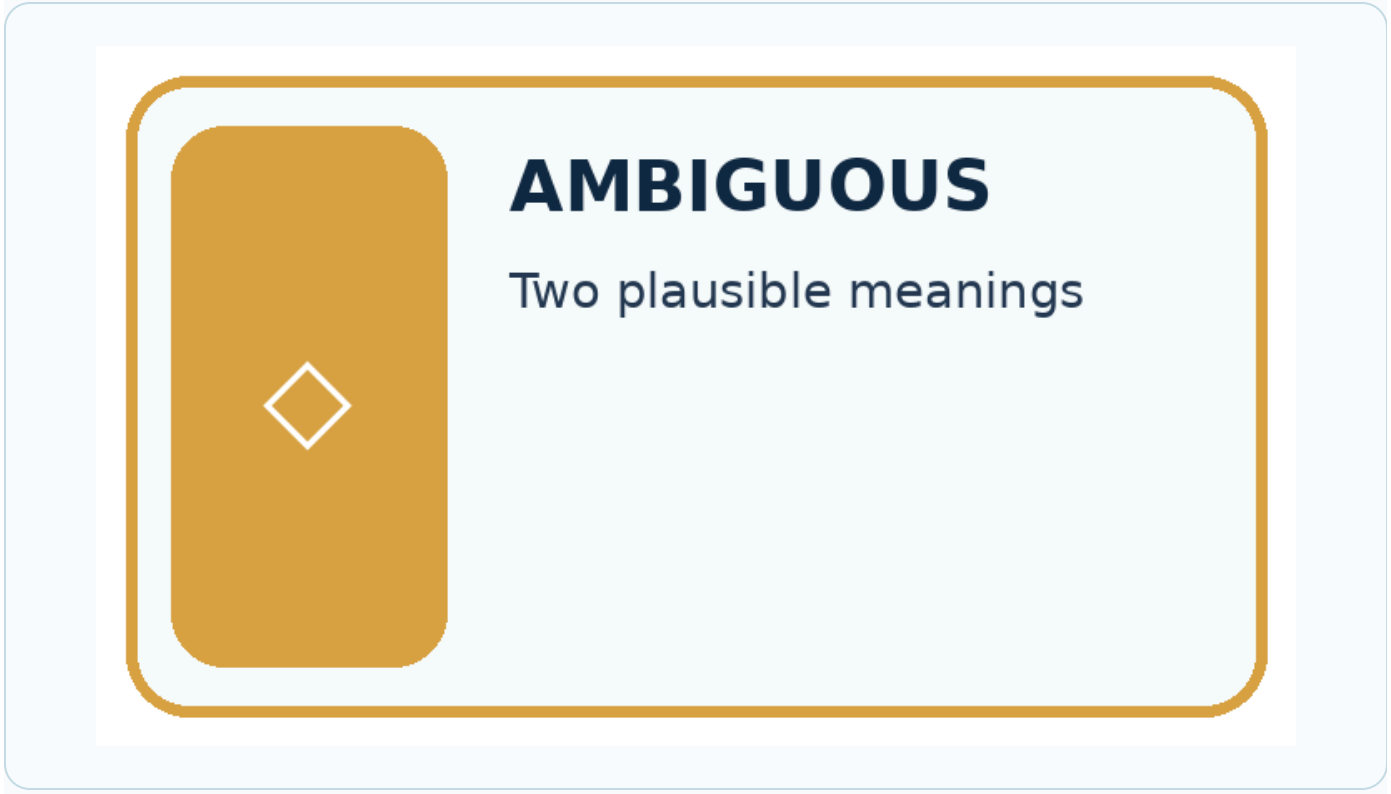
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace a known thirteen/thirty mismatch

> Keep the separate unresolved fifteen/fifty inventory case distinct.

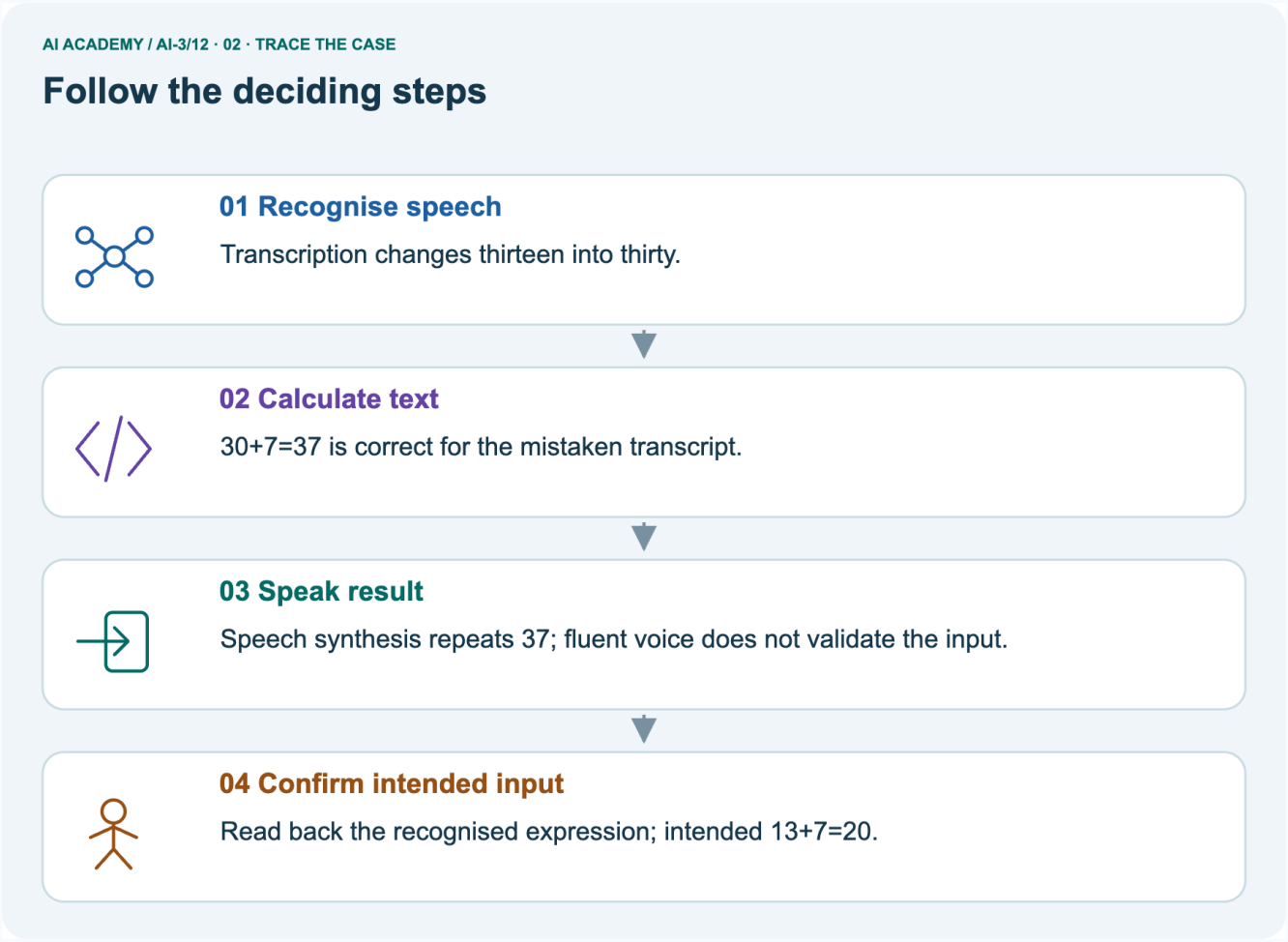
Read the known phrase, transcript and spoken result

A fictional learner dictates a calculation to a voice assistant.

Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37.

Synthetic voice says “thirty-seven”. The example supplies ground truth; no real audio was analysed.

Trace the earliest mismatch and later correct processing



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Recognise speech	Transcription changes thirteen into thirty.
2. Calculate text	30+7=37 is correct for the mistaken transcript.
3. Speak result	Speech synthesis repeats 37; fluent voice does not validate the input.
4. Confirm intended input	Read back the recognised expression; intended 13+7=20.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

The first error occurs before arithmetic. Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

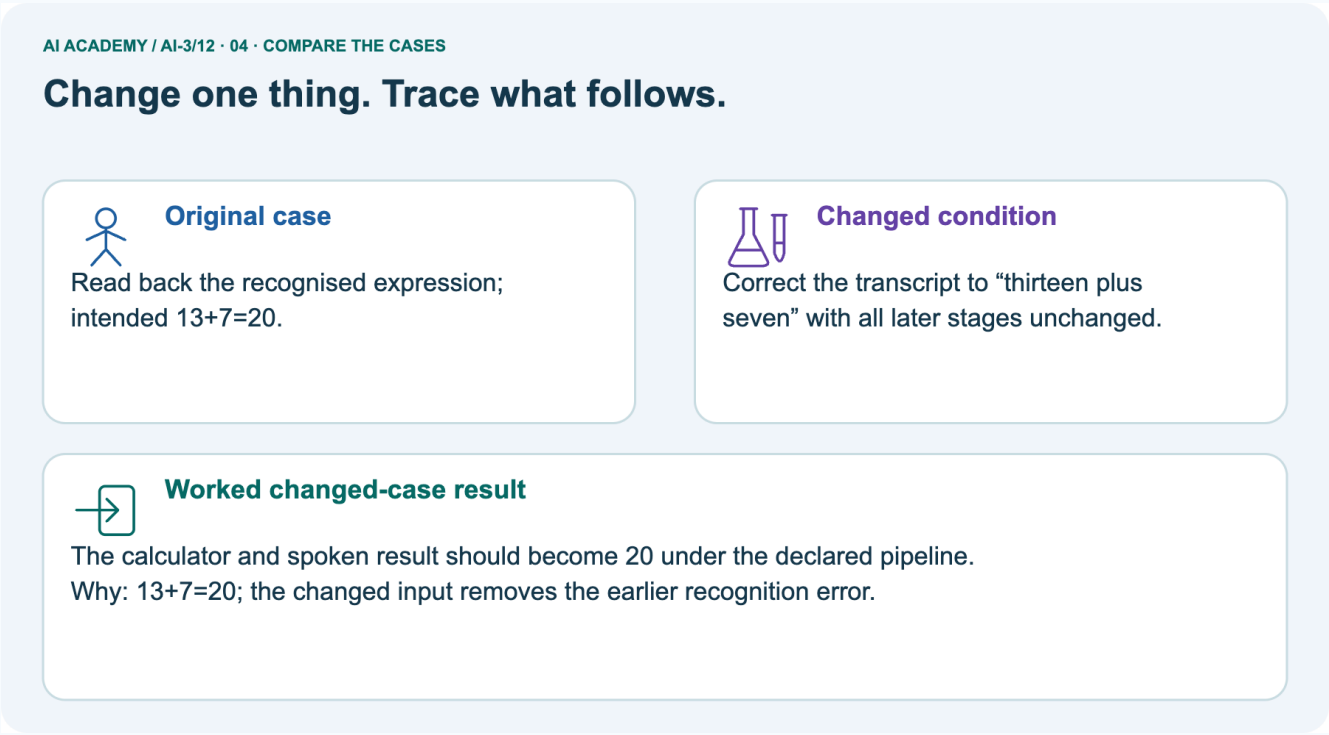
Clarify the counterfactual change

Here the known phrase is supplied as classroom ground truth; no actual recording or recogniser is tested. The parsed expression follows the supplied transcript. Correcting the transcript while keeping later stages unchanged means using the same interpretation, arithmetic and speech-rendering rules with new input; it does not freeze their old output text. The corrected13+7 result is20.

Predict correcting the transcript

- Change one thing: Correct the transcript to “thirteen plus seven” with all later stages unchanged. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State limits on recogniser-performance claims

- These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

- Read the labelled picture: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

▼ Separate review · Unavailable audio versus inventory

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

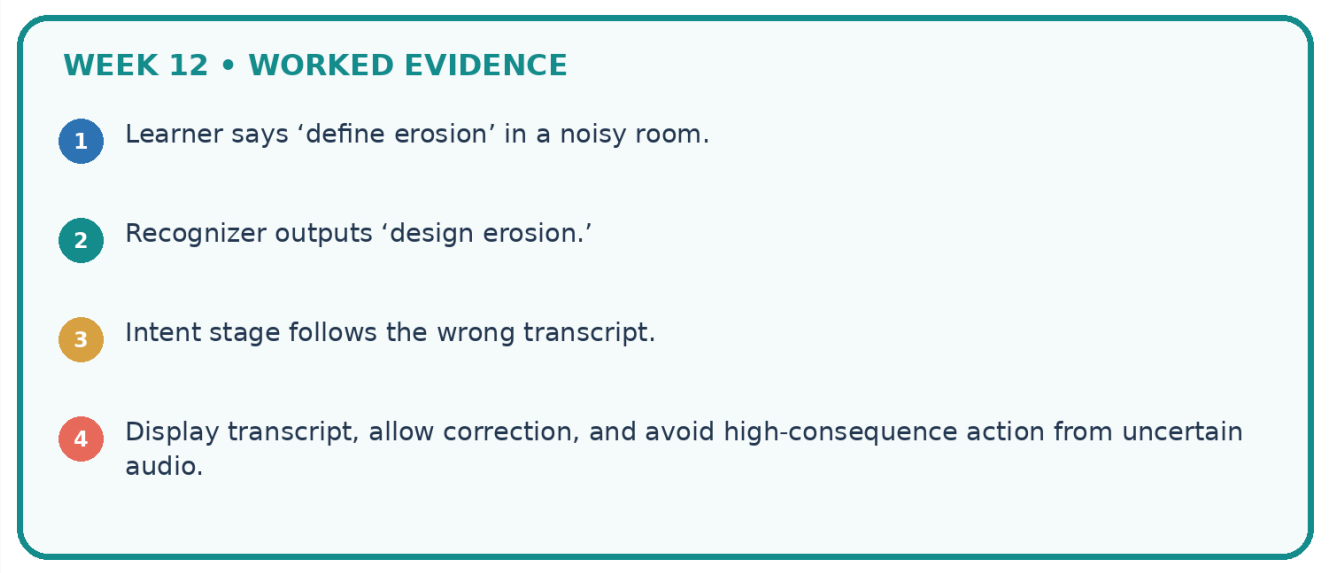
There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Learner says ‘define erosion’ in a noisy room.

2. Recognizer **output**s ‘design erosion.’

3. **Intent** stage follows the wrong transcript.

4. Display transcript, allow correction, and avoid high-consequence action from uncertain audio.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

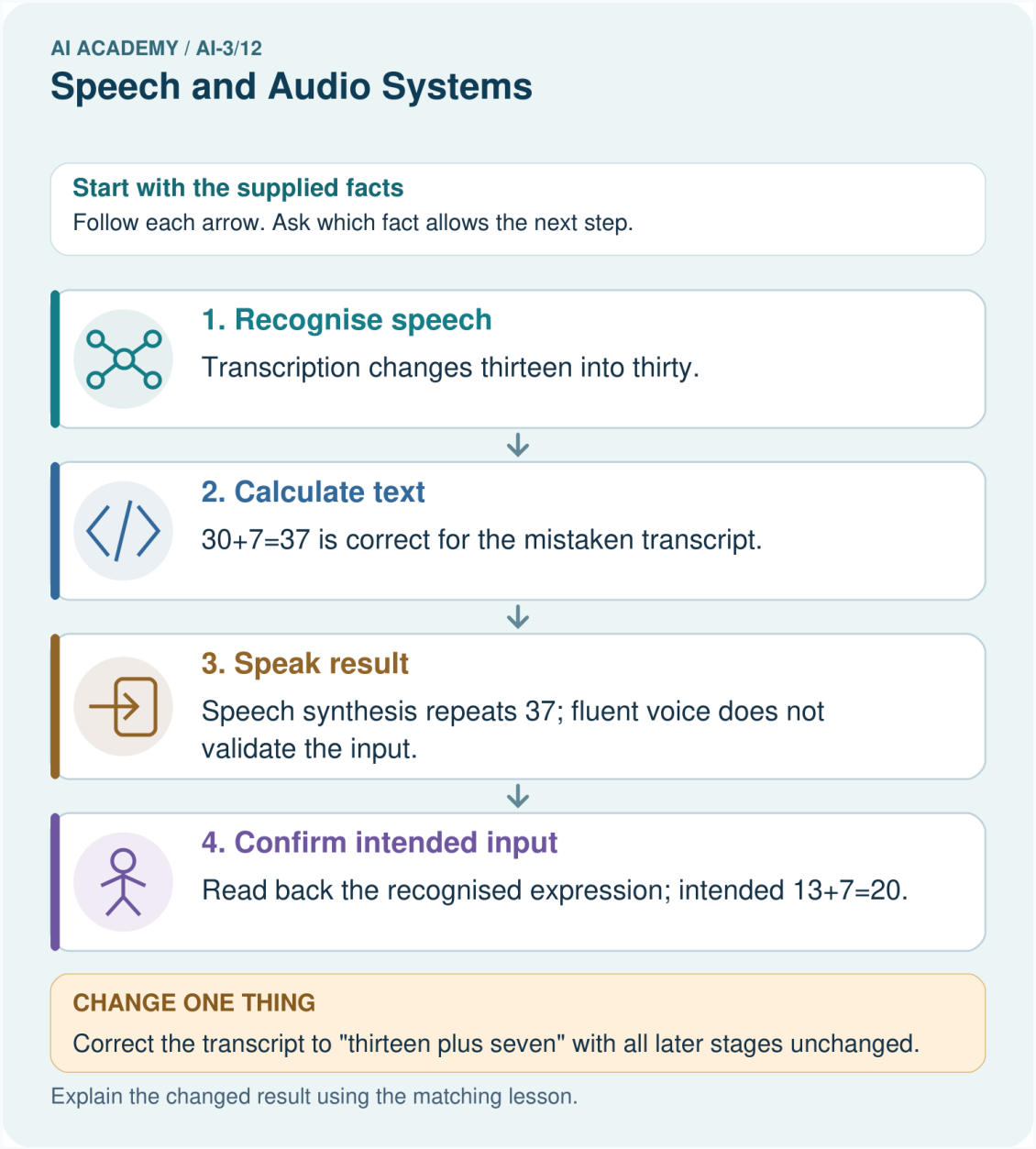
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Speech and Audio Systems

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the wrong inference from a correct downstream output

> Check the input to a stage as well as its output.

Repair the fluent-voice argument

Claim to repair: The voice said37clearly, so the learner must have asked30+7. The supplied reference utterance is13+7, while the transcript says30+7. Correct arithmetic and fluent synthesis can preserve that earlier error. In the separate boxes case the original audio is unavailable, so a conflict with inventory cannot by itself locate a recognition error. State the first supported mismatch and what remains unknown.

Propose a stage-specific check

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A wrong spoken response means the language model caused the first error. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the unresolved conflict with less help

> Do not invent a recording or authority rule.

The stage/authority evidence boundary

A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Complete the separate fifteen/fifty W2 record

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Speech pipelines and multimodal **evidence**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair an all-future-cases claim

- Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the known thirteen/thirty picture

- Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

Explain the picture

My explanation using a pictured step and the given facts

Reserve the independent nineteen/ninety case

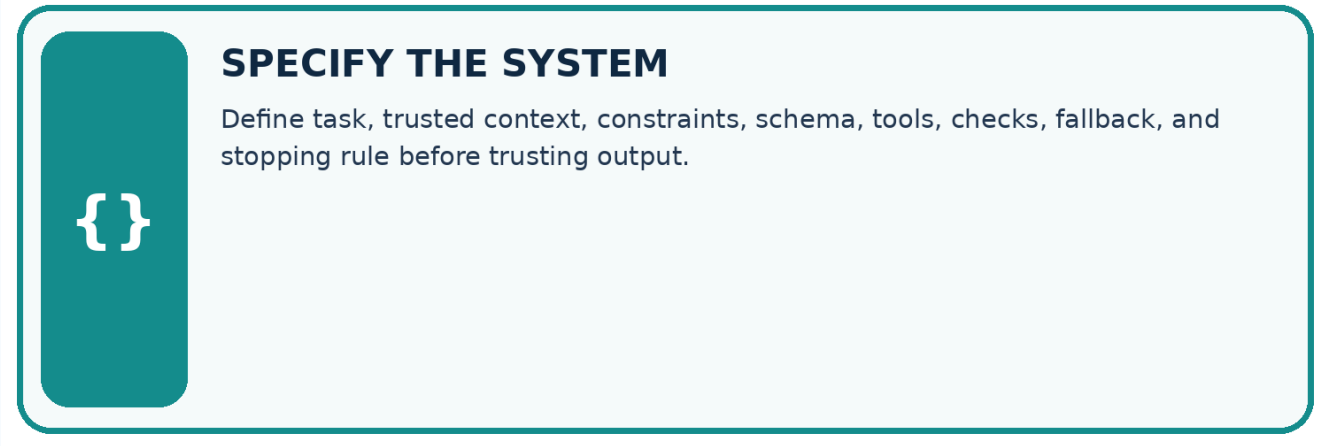
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Diagnose four complete stage records

> Parsed operands are supplied so a calculation error can be distinguished from interpretation.

The four supplied cards

Card · Known source · Transcript · Parsed input · Calculator · Spoken/final output

> **Card:** A
>
> **Known source:** thirteen plus seven (given reference)
>
> **Transcript:** thirty plus seven
>
> **Parsed input:** 30+7
>
> **Calculator:** 37
>
> **Spoken/final output:** thirty-seven

> **Card:** B
>
> **Known source:** thirteen plus seven (given reference)
>
> **Transcript:** thirteen plus seven
>
> **Parsed input:** 13+7
>
> **Calculator:** 27
>
> **Spoken/final output:** twenty-seven

> **Card:** C
>
> **Known source:** thirteen plus seven (given reference)
>
> **Transcript:** thirteen plus seven
>
> **Parsed input:** 13+7
>
> **Calculator:** 20
>
> **Spoken/final output:** twenty-seven

> **Card:** D
>
> **Known source:** Original audio unavailable; inventory says50boxes; no source priority supplied
>
> **Transcript:** fifteen boxes
>
> **Parsed input:** No arithmetic task

> **Calculator:** not applicable
>
> **Spoken/final output:** fifteen boxes for certain

Trace the first mismatch and preserve limits

1. Four complete paper trace cards: for A–C compare the supplied reference phrase with transcript, then transcript with parsed input, independent arithmetic with calculator output, and intended number with spoken rendering. Record the first supported mismatch, later stages that agree with their inputs, and one revealing check.

   In D preserve the transcript/inventory conflict and absent replay; the certainty claim is unsupported, but the first technical stage failure cannot be identified from these facts. Neither modality has a declared priority. Ask for the missing evidence instead of choosing a convenient quantity. These are constructed records, not real audio analysis, measured accent/noise performance or actual speech synthesis.

   Keep the independent nineteen/ninety trace withheld.

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

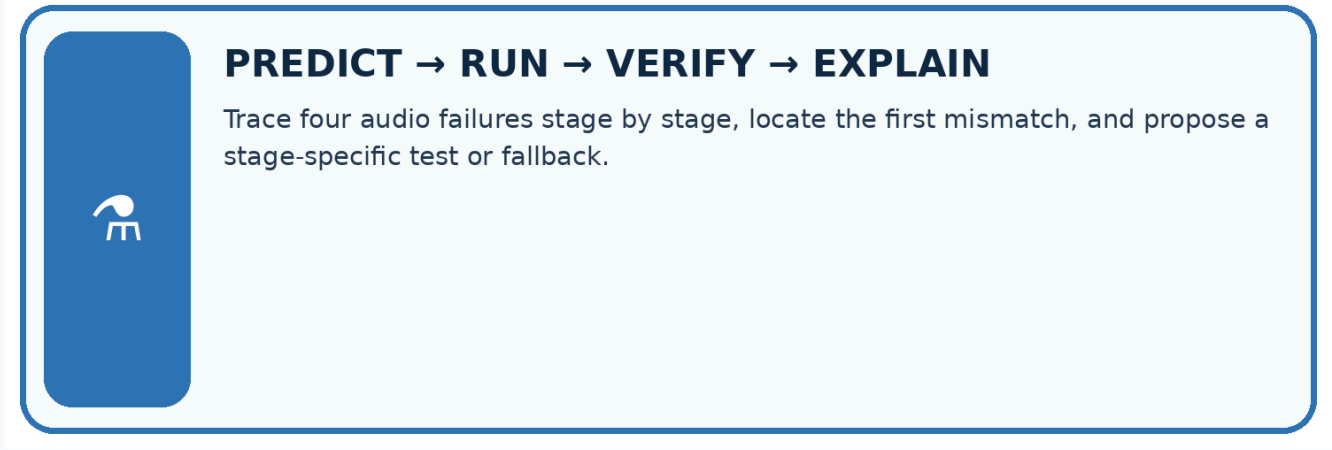
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the four complete paper stage records, including parsed operands. Identify the first supported recognition, calculation or spoken-output mismatch; keep the unavailable-audio inventory conflict unresolved. No recording, live recogniser or actual synthesis is tested.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: locate the fresh nineteen/ninety mismatch

> Distinguish the first wrong representation from correct arithmetic on it.

Independent record and scope

Known audio says “nineteen minus six”. **Transcription** says “ninety minus six”; the calculator returns84. Locate the first error and propose a check.

The independent task supplies the known phrase and transcript as classroom facts, not actual audio. Treat the parsed operands as90and6 with subtraction, as stated by the calculator trace. Check its arithmetic against that received input and separately recover the intended reference expression. Do not infer real recogniser accuracy or claim a replay was performed.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 12 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A recording says fifteen, but the transcript says fifty. The summary repeats fifty. Where did the error enter, and what should be checked first?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Show transcripts and require confirmation before saving memory or using tools.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Make the project's recognised question inspectable

> Check important numbers before downstream decisions.

This week's concrete contribution

For the Verified Study Buddy, show the recognised question and parsed numbers/operation before using them in important calculations or actions. Preserve the confirmed input and stage evidence needed for the declared task; use teacher-managed fictional records in this lesson. A read-back can expose a misunderstanding, but a fluent confirmation is not itself proof of correctness.

Keep recognition, retrieval, calculation and response claims separately checkable.

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates.

Keep the checkpoint and human choice

THIS WEEK'S CHECKPOINT Show transcripts and require confirmation before saving memory or using tools.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

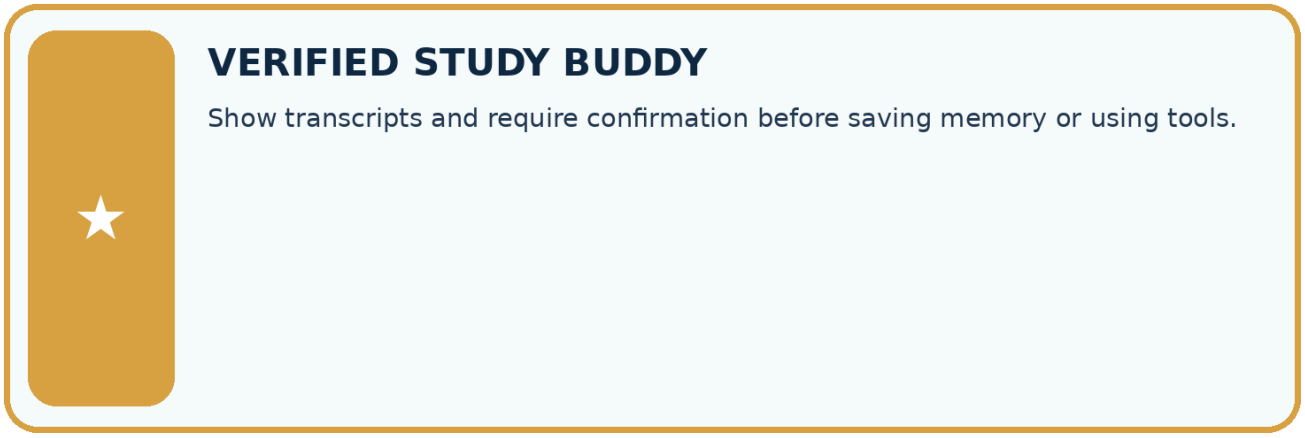
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: name the first supported error and the missing evidence

> A correct later step cannot establish that the original input was right.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
capture	Record a signal, observation or item of information for later use.	
speech recognition	Converting spoken audio into an estimated sequence of words.	
intent	The goal a user is trying to achieve, which may need clarification.	
synthesis	Combining information into a new account or output while preserving its meaning and support.	

Terms for the connected explanation

Word	Meaning
Modality	A form of information such as audio, image or text
Transcription	A text representation of speech

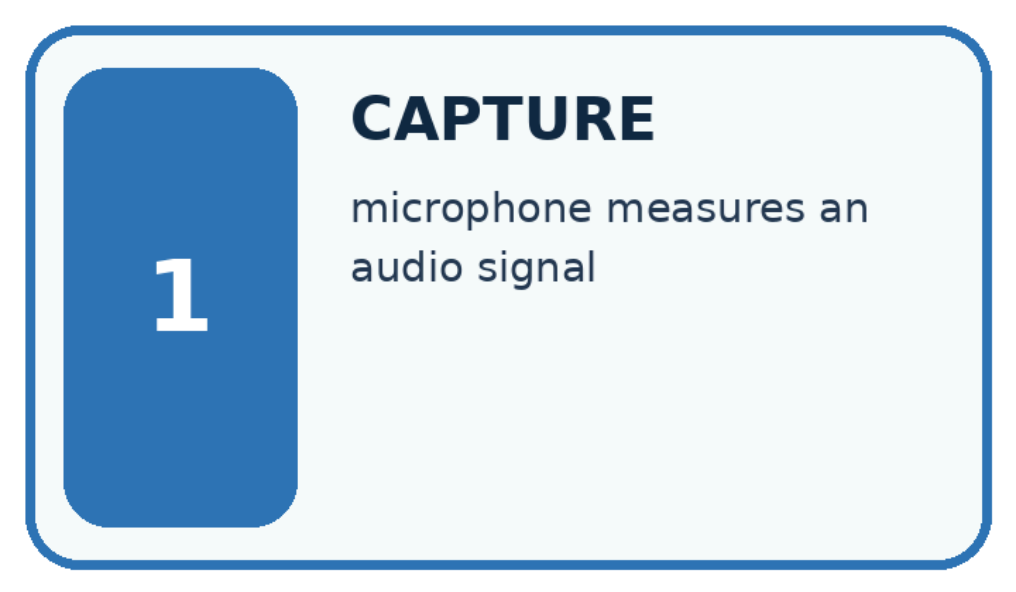
Word	Meaning
Error propagation	An earlier error affecting later processing

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

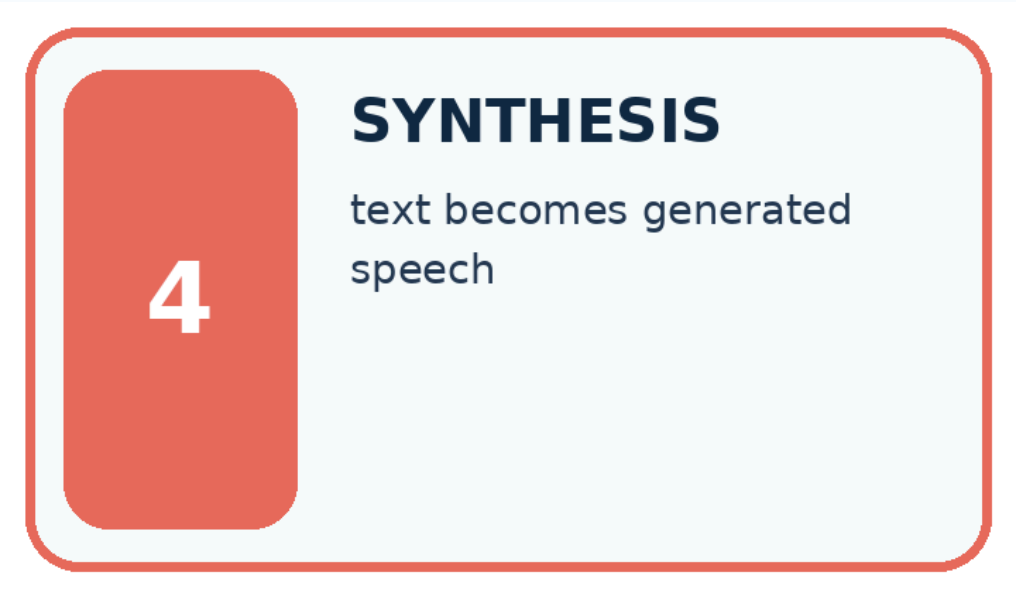
SPEECH RECOGNITIC

signal becomes candidate words

3

INTENT

system’s interpretation of the request



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

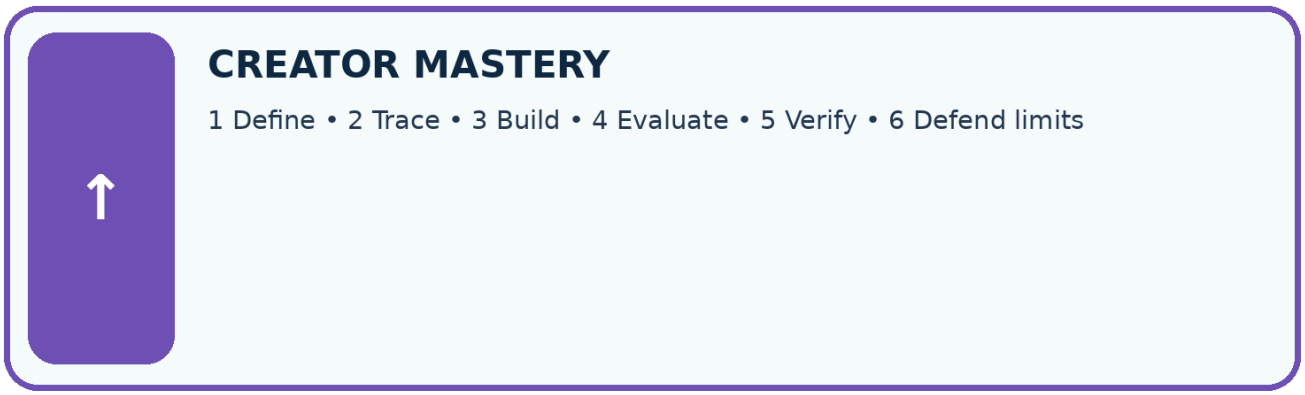
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Diagnose a new calculation trace independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

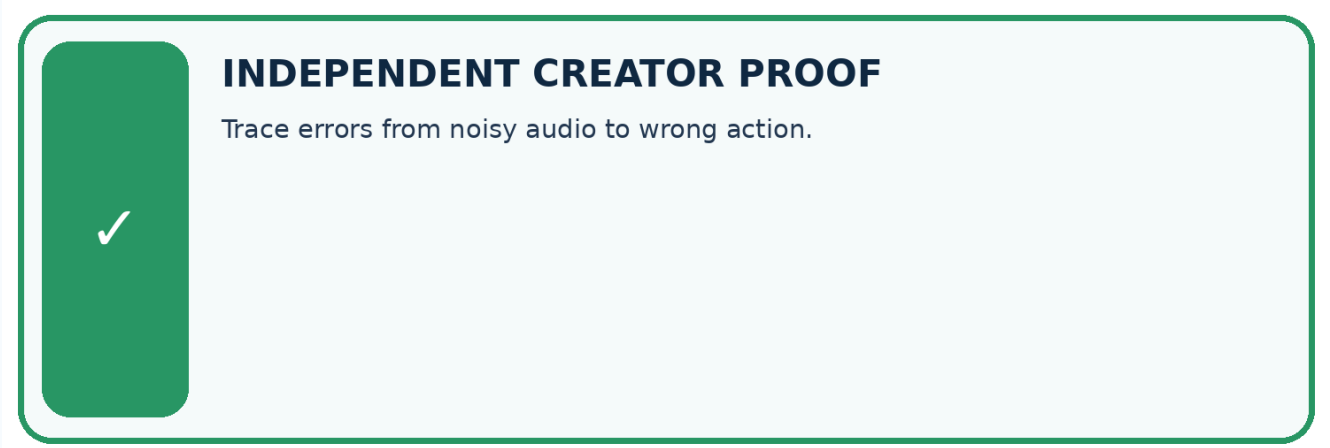
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Diagnose a new calculation trace independently is the selected independent case.



EXPLANATION CHECK Require the first supported mismatch, correct downstream arithmetic, intended result and a revealing confirmation check.

Write or draw your reasoning and evidence.

▼ **Extra practice: Trace a speech error through the system**

REFERENCE · SECTION 13

Extra practice: Trace a speech error through the system

Try the question before opening its worked explanation.

Read the supplied case and question

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

Where did the error enter?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Transcription supplied the wrong number. Confirm the transcript or ask for clarification.
Correct arithmetic cannot repair wrong **input**.

Evidence or stage	Value or result
Speech	Fifteen
Transcript	Fifty, the first error
Calculator	Correct arithmetic using fifty
Result	Wrong for the original request

▼ Earlier wording, alternatives and picture transcripts

Week 12 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Trace a speech error through the system

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

Where did the error enter?

Transcription supplied the wrong number. Confirm the transcript or ask for clarification. Correct arithmetic cannot repair wrong **input**.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of speech pipelines and multimodal evidence	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional learner dictates a calculation to a voice assistant.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional learner dictates a calculation to a voice assistant.
- Which labelled step matches this detail? Read back the recognised expression; intended 13+7=20.
- 1. Confirm intended **input**
- 2. Recognise speech
- 3. Calculate text
- 4. Speak result
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Confirm intended **input** Read back the recognised expression; intended 13+7=20. The first error occurs before arithmetic.

Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

Changed-case question

- Changed-case practice: Correct the transcript to “thirteen plus seven” with all later stages unchanged. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The calculator and spoken result should become 20 under the declared pipeline.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The calculator and spoken result should become 20 under the declared pipeline. 13+7=20; the changed **input** removes the earlier recognition error.

Repair and save your work

Review Visual Composition (ai-3/11). Goal: Check object counts, spatial relations and arrow meaning separately. Short practice: Which checks pass after that single change, and what still needs repair?. Return to this week and explain what you changed.

Review Prompt Evaluation (ai-3/9). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Why is B’s selection defensible here without calling it universally superior?.

Return to this week and explain what you changed.

Review Examples and Rubrics (ai-3/6). Goal: Distinguish a format example from approved factual evidence. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

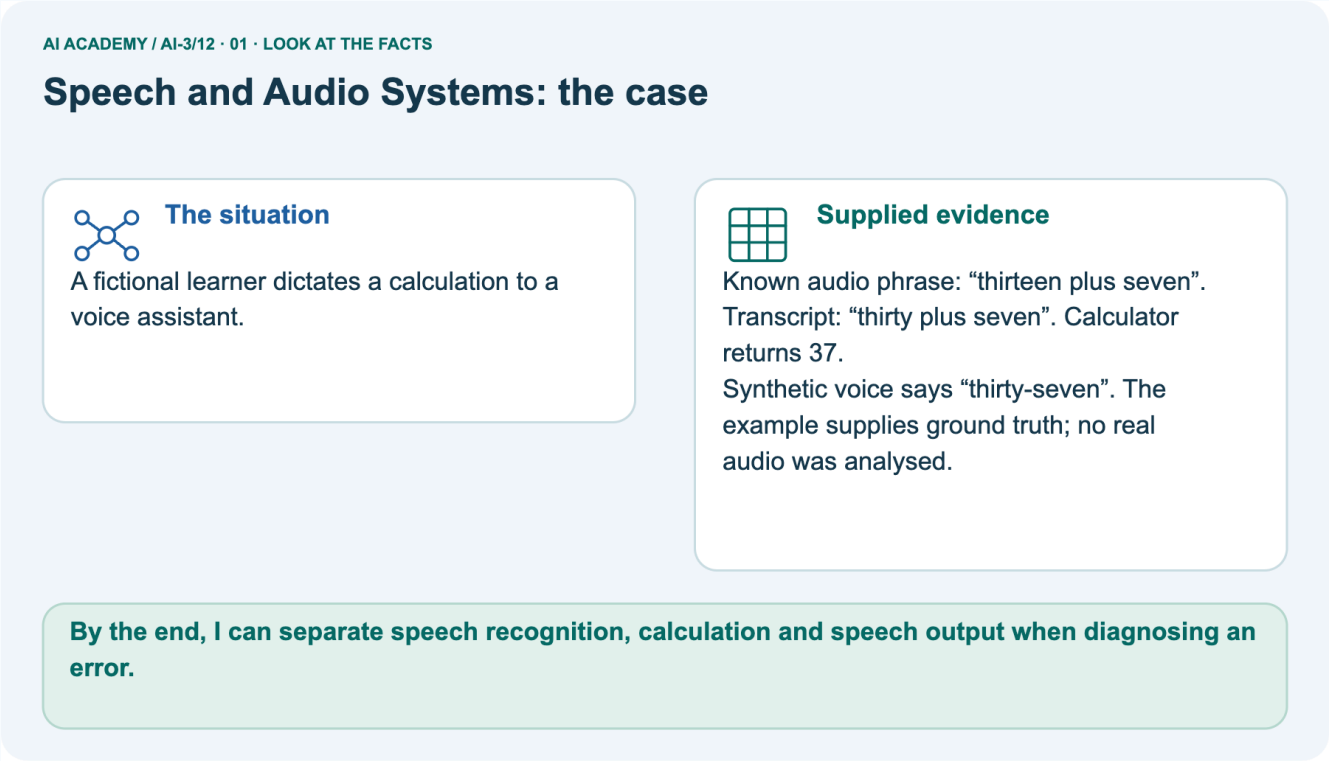
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Speech and Audio Systems: the case



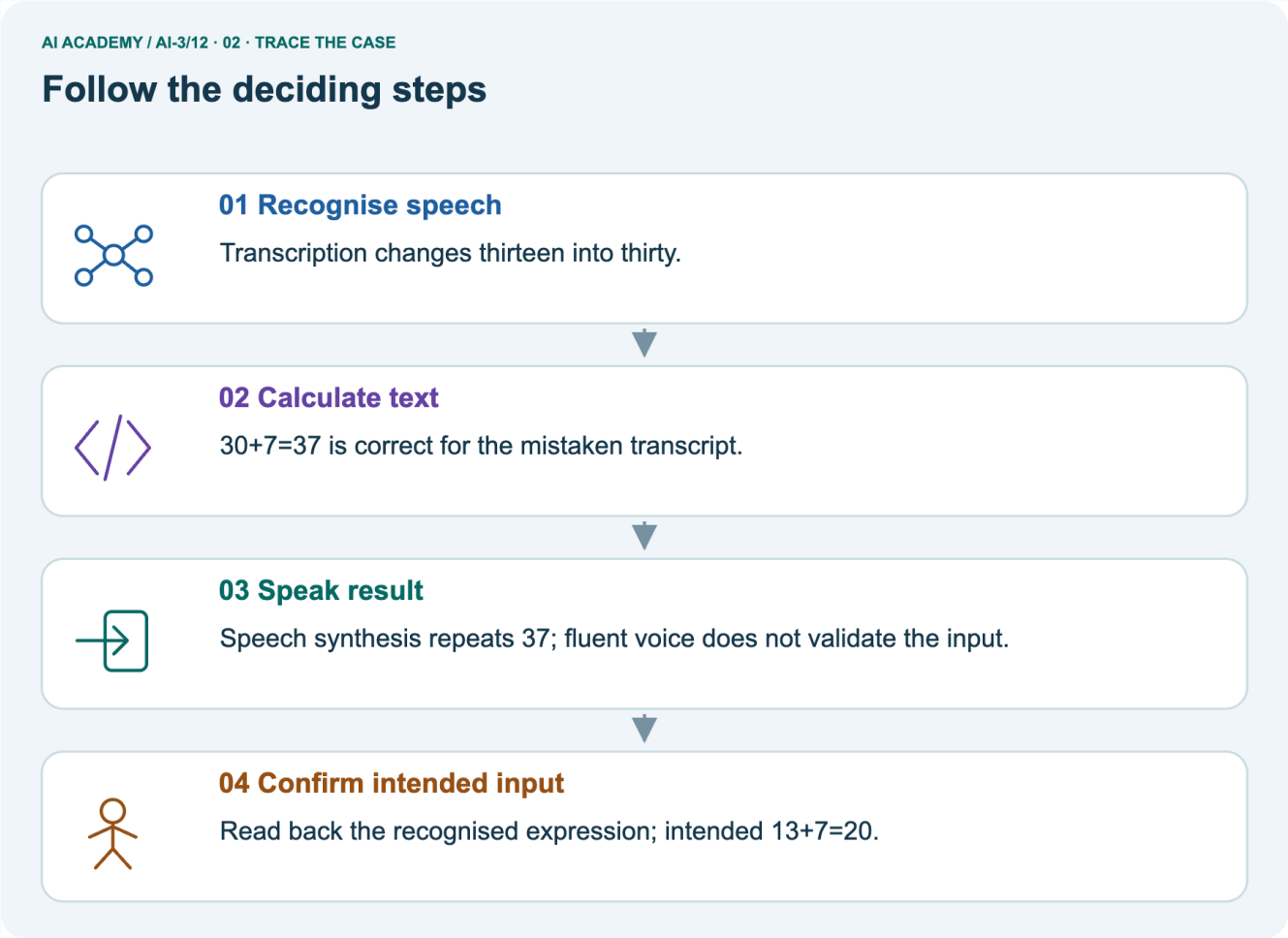
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional learner dictates a calculation to a voice assistant. Supplied facts: Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37. Synthetic voice says “thirty-seven”.

The example supplies ground truth; no real audio was analysed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

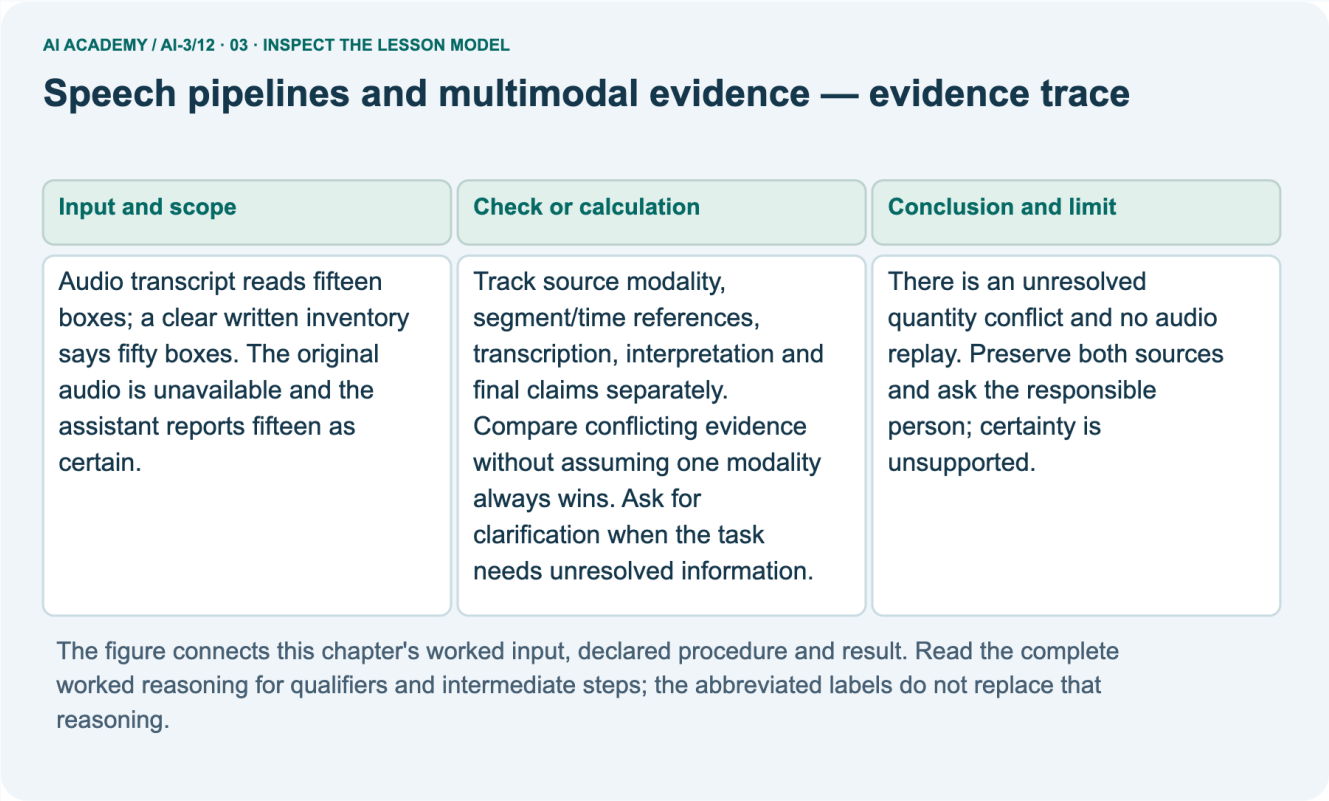


Follow the labelled picture with the matching explanation. Open the vector diagram

Recognise speech: **Transcription** changes thirteen into thirty. → Calculate text: 30+7=37 is correct for the mistaken transcript. → Speak result: Speech **synthesis** repeats 37; fluent voice does not validate the **input**. → Confirm intended **input**: Read back the recognised expression; intended 13+7=20.

Point to each numbered stage and explain which recorded **input** it uses.

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech pipelines and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

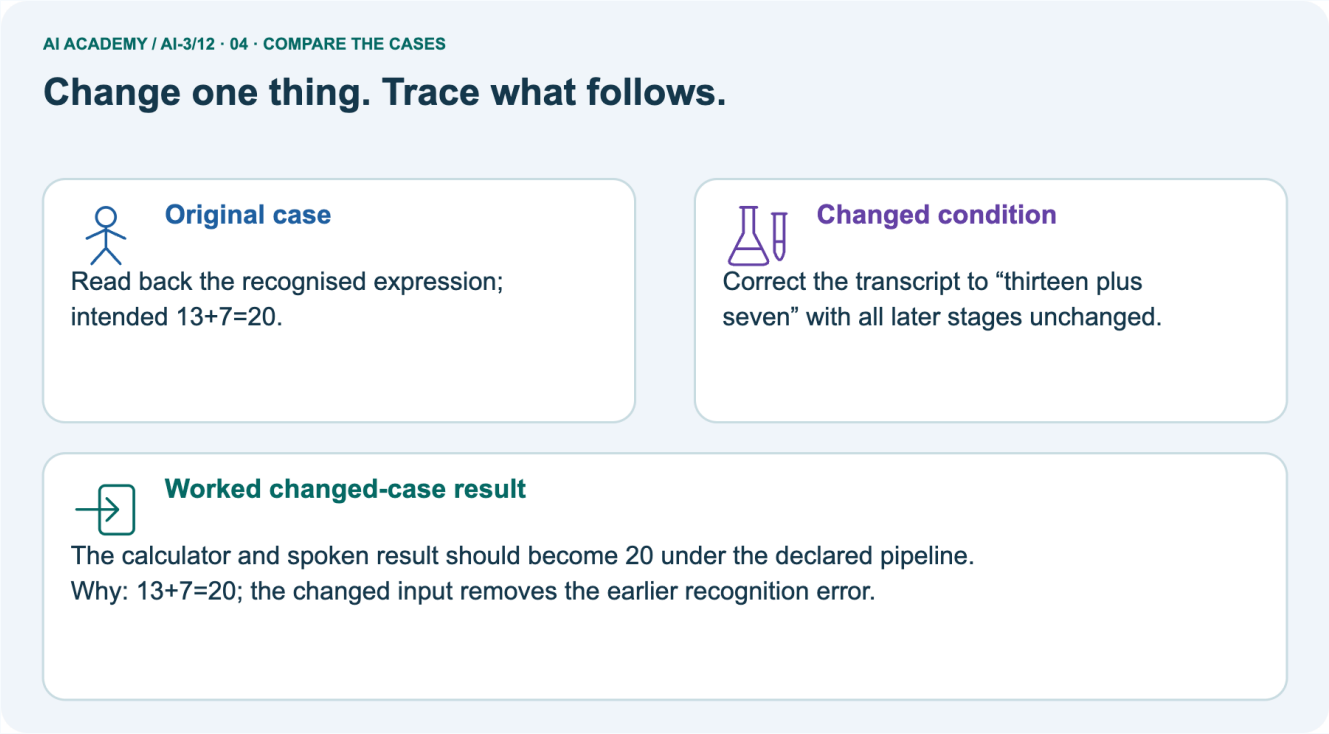
The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity conflict and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



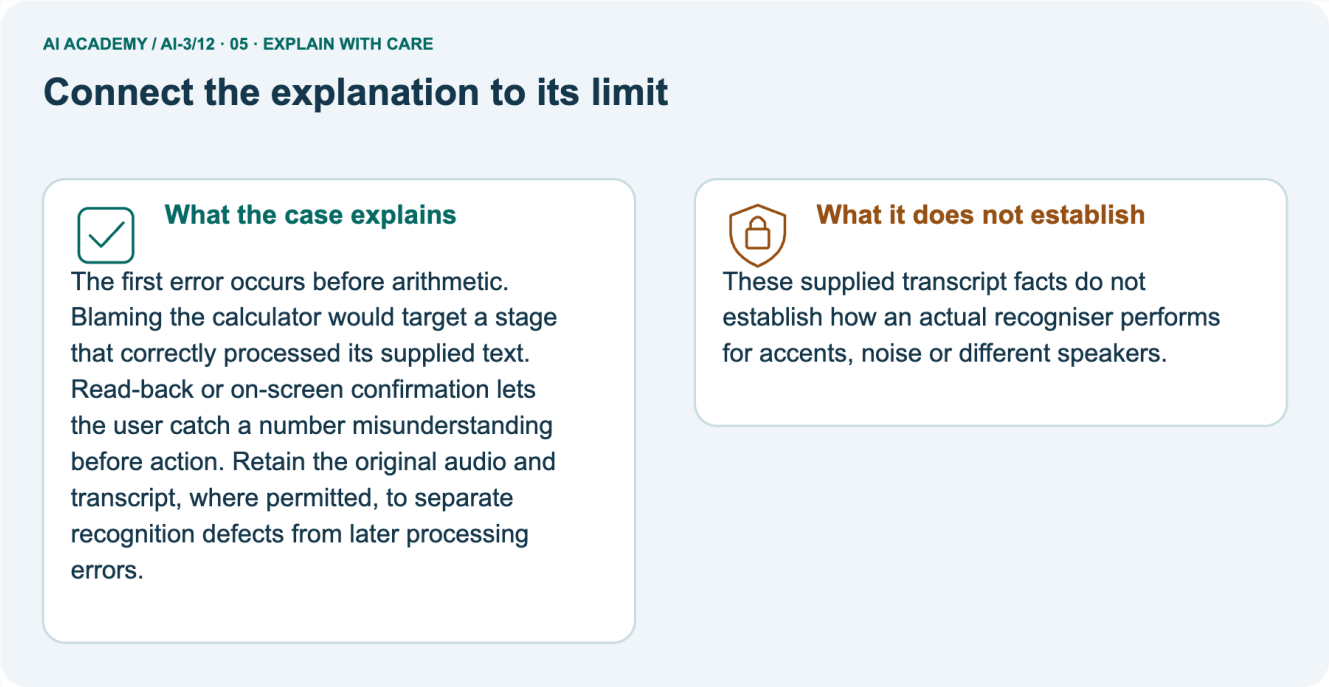
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Read back the recognised expression; intended 13+7=20. Changed condition: Correct the transcript to “thirteen plus seven” with all later stages unchanged.

Expected result: The calculator and spoken result should become 20 under the declared pipeline. Reason: 13+7=20; the changed **input** removes the earlier recognition error.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

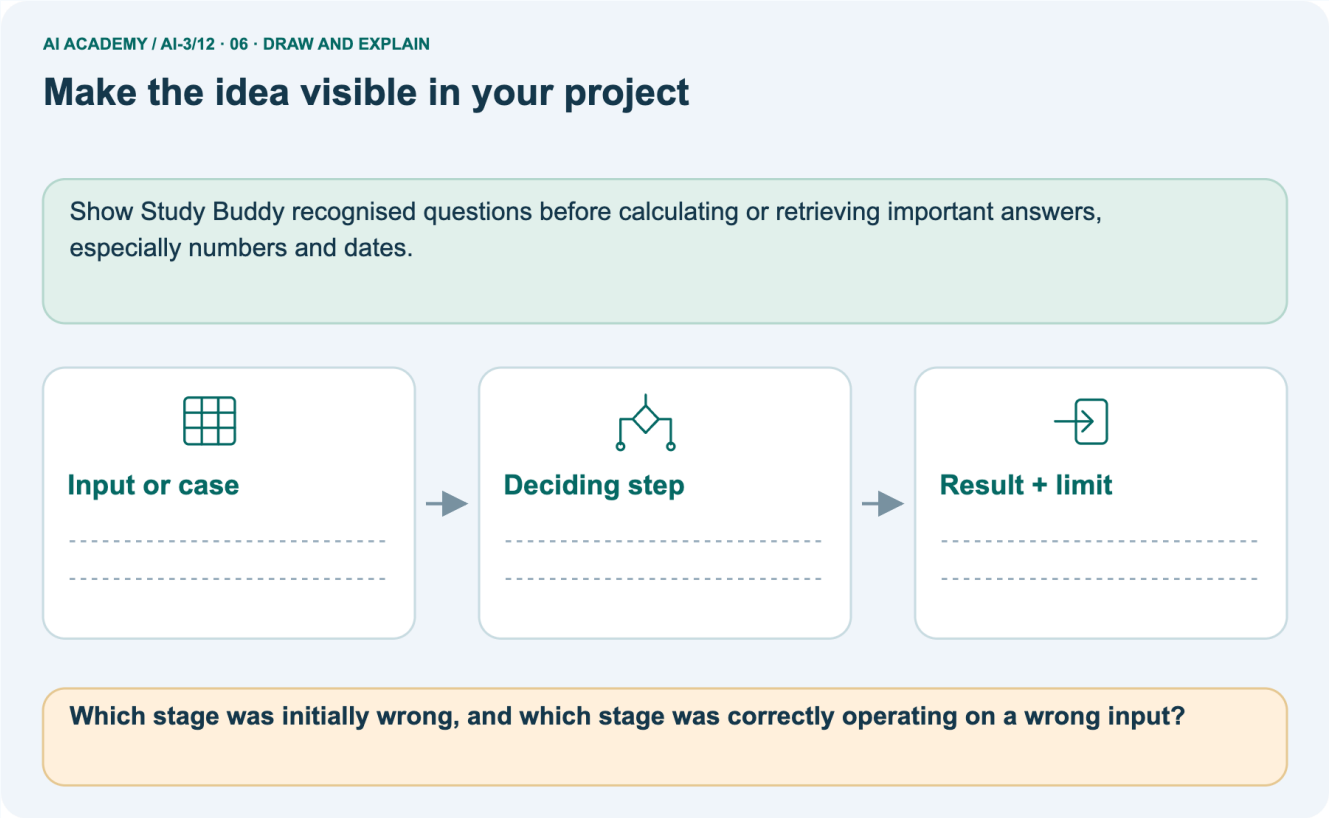
The first error occurs before arithmetic. Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

Limit: These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates. Draw your **input**, deciding step, result and limit.

Supported question: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Speech and Audio Systems: the case

02. Follow the deciding steps

03. Speech pipelines and multimodal evidence — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.